



K.R. MANGALAM UNIVERSITY

Database Management Lab File
(ENCS205)

Bachelor of Technology
in
Computer Science and Engineering

B. Tech CSE (Core)
4 Semester

School of Engineering & Technology

K. R. MANGALAM UNIVERSITY Sohna, Haryana
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Submitted By:	Submitted To:
Vishal Kumar (2401010310)	MS. Mansi Kajal

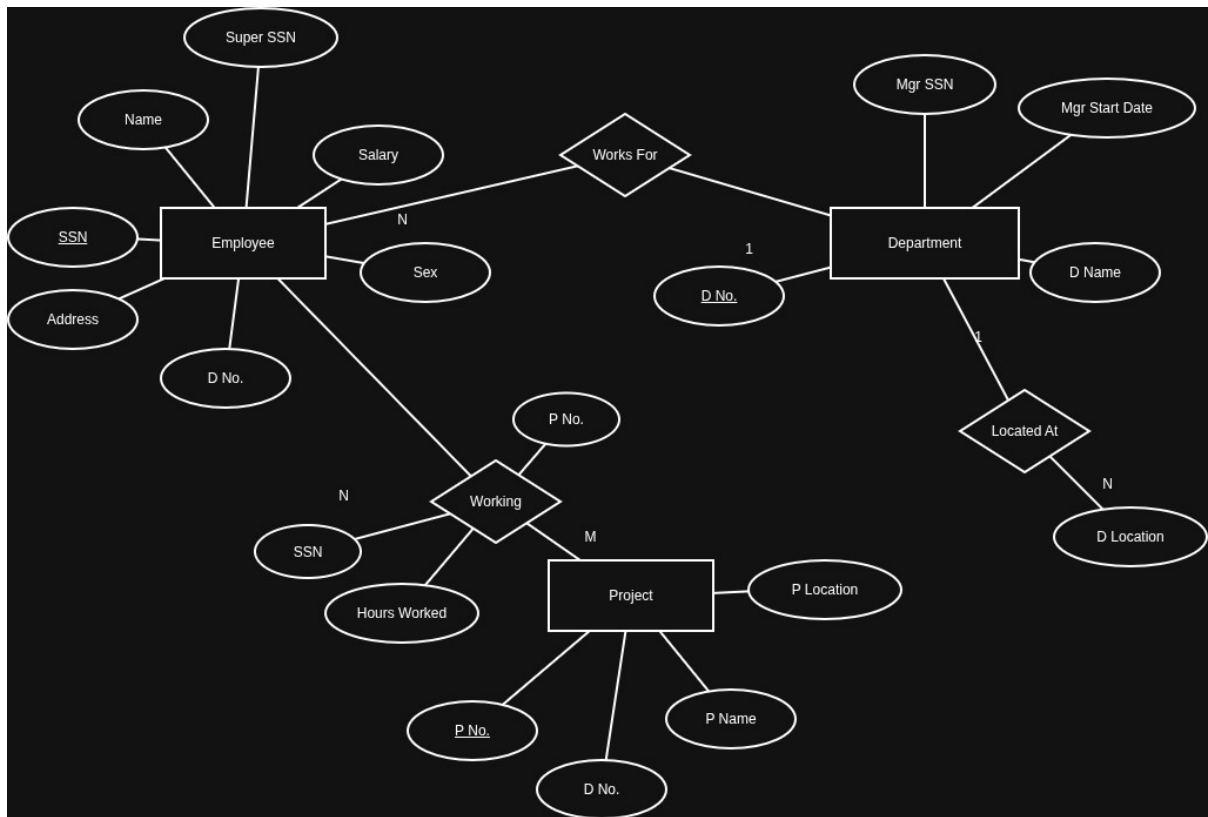
EXPERIMENT-1

Consider following Company databases and draw ER diagram and convert entities and relationships to relation table for a given scenario:

EMPLOYEE (SSN, Name, Address, Sex, Salary, Super SSN, D No.)

DEPARTMENT (D No., D Name, Mgr SSN, Mgr Start Date) DLOCATION (D Location)

PROJECT (P No., P Name, P Location, D No.) WORKING (SSN, P No., Hours Worked)



- The Company Database ER diagram represents the organizational structure of a company. It stores information about employees, departments, projects, and work assignments.
- The main entity is Employee, which contains details of all employees working in the company.
- The attributes include SSN, Name, Address, Sex, Salary, SuperSSN, and DNo.
- SSN uniquely identifies each employee.
- Therefore, SSN is the primary key.
- SuperSSN refers to the supervisor of an employee.
- This creates a recursive relationship where an employee can supervise other employees.
- DNo connects the employee to a department and acts as a foreign key.
- The Department entity stores department information.
- Its attributes include DNo, DName, MgrSSN, and MgrStartDate.
- DNo uniquely identifies each department.
- Hence, it is the primary key.
- MgrSSN represents the manager of the department and references the Employee entity.

- The Works For relationship connects Employee and Department.
 - One department can have many employees.
 - Each employee works for only one department.
 - Therefore, it is a one-to-many relationship.
- The Project entity stores project details handled by departments.
- Its attributes include PNo, PName, PLocation, and DNo.
 - PNo uniquely identifies each project and is the primary key.
 - DNo links the project to a department.
- The Works_On relationship connects Employee and Project.
- It includes SSN, PNo, and Hours.
 - Hours shows the number of hours worked on a project.
 - Since many employees can work on many projects, this is a many-to-many relationship.
 - Hence, Works_On becomes an associative relation with a composite primary key.
- The DLocation entity stores the locations of departments.
- Its attributes are DNo and DLoc.
 - A department may have multiple locations.
 - Therefore, this forms a one-to-many relationship.
- This ER model ensures proper organization of company data, avoids duplication, and maintains integrity between related tables.
- It helps management track employees, projects, departments, and working hours efficiently.